How a Predictive State Observer Can Self-Adapt Its Sensory Prediction-Error Correction Gain: Closed-Loop Evidence from a Muscle-Driven Reaching Task

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Abstract

We ask whether a forward-model-based predictive state observer — an internal forward model in the cybernetic sense — can adapt its sensory prediction-error correction gain from signals the agent itself generates online (innovation history and per-episode reaching outcome), rather than from per-condition oracle labels. On a 34-muscle MyoSuite reaching arm, a predictive observer over a residual forward model uses a within-trial reliability-to-gain logistic and an across-trial outcome-driven update of its meta-parameters $\boldsymbol{\beta}$ via a gradient-free two-point stochastic approximation rule, deployed in closed loop with a joint-space PD controller as a state-estimation-sensitive probe. This yields a mechanistic decomposition of the closed-loop correction-gain geometry, which is dominated by forward-model error and bias, not sensory reliability: a regression of per-cell $\boldsymbol{\Delta}\textbf{outcome}(\boldsymbol{K}{=}\mathbf{1}, \boldsymbol{K}{=}\mathbf{0})$ across four forward-model qualities reaches $\boldsymbol{R^2 = 0.95}$, against $\boldsymbol{R^2 = 0.17}$ from reliability variance alone. At $\boldsymbol{n}{=}\mathbf{200}$ episodes per cell, default within-trial reliability trails the $\boldsymbol{K}{=}\mathbf{0}$ baseline by **22** cm at delay-**18** cells and **35** cm at delay-**0** cells. Outcome-driven optimisation of a single global $\boldsymbol{\beta}$ modestly improves delay-**18** by $\approx$**5** cm but yields $\approx$**0** improvement at delay-**0**, directly exhibiting the parameterisation ceiling of the reliability-to-gain logistic. A **30**-parameter diagonal field-wise linear adapter that maps z-scored log innovation statistics to $\boldsymbol{\beta}$ further reduces delay-**18** error to **0.40** m, within **2.0** cm of $\boldsymbol{K}{=}\mathbf{0}$, while delay-**0** cells remain unchanged. Together, the results identify two distinct ceilings: a *context aggregation ceiling* that agent-available innovation statistics can substantially close at delay-**18**, and a *parameterisation ceiling* of the reliability-to-gain logistic that they cannot.

Keywords: Forward model, Predictive state observer, Sensory prediction-error correction, Adaptive correction gain, Closed-loop motor control, Muscle-driven reaching

1 Introduction

State estimation in muscle-driven manipulation faces three coupled challenges: sensorimotor delay, observation noise, and signal-dependent motor noise. The internal-model view of biological motor control answers these with a *forward-model-based predictive state observer*: an internal forward model uses efference copy to predict the next state, and the observer corrects that prediction by a scalar weight $K \in [0, 1]$ on the sensory prediction error Bernstein (1967); Wolpert and Ghahramani (2000). $K{=}0$ trusts the forward prediction (efference copy); K=1 trusts the sensor; intermediate K implements a sensory-prediction-error correction. The innovation-update form is structurally familiar from Kalman filtering, where the gain is derived from propagated covariance estimates Kalman (1960); Mehra (1970); Anderson and Moore (1979); we do *not* propagate covariances and do not claim to implement a full Kalman filter (Sec. 3.3). Instead we treat K as a scalar (or field-wise vector) that controls how strongly sensory prediction error corrects the current state estimate, and ask: *can the agent learn to set this gain from signals it generates online — innovation history and per-episode task outcome — rather than from the per-condition K-sweep oracle labels that supervised gain predictors use at training time (Appendix C)?*

This question is biologically motivated. The nervous system does not have access to a swept oracle $K^\star$ across noise and delay regimes; it has access to its own prediction errors and, after moving, to whether the limb arrived where intended. We therefore propose a *two-layer reliability-adaptive observer*. *Within-trial*, a per-channel exponentially-weighted variance of squared innovation gives a running reliability estimate $r_f(t)$, and a logistic $K_f(t)=\sigma(\beta_{0,f}+\beta_{1,f}\log r_f(t))$ maps it to a field-wise correction gain on the qpos, qvel, act, tip_pos, and reach_err sensory channels. *Across-trial*, Simultaneous Perturbation Stochastic Approximation (SPSA) Spall (1992) updates the meta-parameters $\beta=\{\beta_{0,f},\beta_{1,f}\}$ from the per-episode reaching outcome. Neither layer accesses the optimal K, the true state, or a swept oracle; both use only signals the agent generates during reaching. The resulting object is a computational internal-model loop in the cybernetic sense: efference-copy-based prediction is corrected by delayed sensory prediction error, while trial-level outcome reshapes the reliability-to-gain rule.

We instantiate this observer on a 34-muscle MyoSuite reaching arm Caggiano et al. (2022) running in the MuJoCo simulator Todorov et al. (2012) with a residual-MLP forward model trained under H-step rollout supervision (H=8) and a fixed-lag predictive state observer that handles configurable observation delay ($d \in \{0, 18\}$ steps) and configurable per-field Gaussian noise. A separate joint-space PD controller with a one-shot inverse-kinematics target then *deploys* the observer in closed loop. As a diagnostic upper bound, we also retain an offline-oracle-supervised correction-gain predictor (Appendix C), which uses a per-condition closed-loop K-sweep to define $K^\star_{\mathrm{cl}}$ and is the strongest non-agent-available baseline available in this setting.

Our contributions are summarised in five claims.

- **C1.** *Default within-trial reliability adaptation is task-misaligned.* With $\beta_{0,f}=0$, $\beta_{1,f}=0.5$, the within-trial rule produces field-heterogeneous intermediate-to- high gains rather than the task-optimal K=0 that the H=8 forward model demands at every focused-grid cell. Across n=200 episodes per cell, default reliability trails K=0 by 22 cm at delay-18 cells and 35 cm at delay-0 cells, with paired 95% bootstrap CIs that exclude zero in every cell (Fig. 2).
- **C2.** *Single-cell across-trial SPSA closes most of the gap on its training cell.* Outcome-driven SPSA on (σ=none, d=18) closes $\sim$ 62% of the default-reliability vs. K=0 gap at n=200 deployment (residual $\sim$9 cm to K=0), with the smoothed training trajectory reaching the original 10-episode K=0 baseline estimate. (Convergence trajectory uses a 10-episode rolling outcome estimate; see Fig. 3.)
- **C3.** *The closed-loop correction-gain geometry is shaped by forward-model error and bias, not by sensory reliability.* A regression of per-cell Δoutcome(K=1, K=0) against agent-available reliability variance vs. forward-model h=10 rollout MSE and bias gives $R^2=0.17$ for reliability alone and $R^2=0.95$ when forward-model features are added (Fig. 5). Sweeping forward-model quality $H \in \{1, 4, 8\}$ plus undertrained H=8 confirms the prediction: $K^\star_{\mathrm{cl}}$ shifts visibly with model quality, with K=1 catching up at delay-18 cells under degraded models (Fig. 6). Cell heterogeneity therefore reflects forward-model accuracy variation across noise/delay operating points, not sensory reliability variation.
- **C4.** *Two distinct ceilings limit reliability-adaptive observers; one can be broken from agent-available signals alone.* A single global β saturates because it aggregates cells with different correction-gain geometries (*context aggregation ceiling*). A per-cell SPSA diagnostic (10-episode deployment per cell; Sec. 4.5.3) resolves

this at delay-18 cells but *fails* at all three delay-0 cells, because the reliability-to-gain logistic $K_f = \sigma(\beta_{0,f} + \beta_{1,f} \log r_f)$ cannot represent the sharp K=0 regime that delay-0 cells require (*parameterisation ceiling*). At n=200 deployment, a 30-parameter diagonal field-wise linear adapter on z-scored log innovation statistics breaks the context aggregation ceiling cleanly: deployed 6-cell-mean min-tip falls from 0.64 m (global SPSA) to 0.56 m, with delay-18 cells reaching 0.40 m, within 2.0 cm of K=0 (Fig. 7). The parameterisation ceiling at delay-0 persists as predicted.

- **C5.** *Task transfer confirms the geometry dependence.* On the variable-target `ReachRandom` variant $K^\star_{\text{cl}}$ is multimodal $\{0, 0.25, 1\}$ across cells (Appendix F); the ReachFixed-trained β trails K=0 at every delay-18 cell by 4–5 cm, while default within-trial reliability matches or beats K=0 at 2 of 3 delay-18 cells.

Together, C1–C5 yield a mechanistic decomposition: closed-loop correction gain is shaped primarily by forward-model error/bias, then by closed-loop utility, and only marginally by sensory reliability; a single global β saturates due to context aggregation but agent-available innovation statistics suffice to break that ceiling; the residual gap on delay-0 cells is a property of the reliability-to-gain logistic, not of the information available to the agent. Appendix C reports an offline-oracle-supervised correction-gain predictor as the strongest non-agent-available baseline; it requires per-condition K-sweep labels and so is reported only as an upper-bound diagnostic.

The rest of the paper is organised as follows. Section 2 positions our work against prior literature on adaptive correction gains and Kalman-style updates, forward models in motor control, learned dynamics, outcome-driven adaptation, and muscle-driven simulation. Section 3 describes the task, the forward model, the fixed-lag predictive state observer, the reliability-adaptive correction-gain rule, the across-trial SPSA loop, the controller, and the evaluation pipeline. Section 4 reports the five claims with the corresponding figures. Section 5 discusses the trade-offs and the implications of agent-available correction-gain adaptation for biological motor control models.

2 Related Work

Adaptive correction gains and Kalman-style updates. The Kalman filter Kalman (1960) and its smoothing counterpart Rauch et al. (1965) are the textbook framework for combining a state-space model with noisy observations under known noise covariances. When the covariances are unknown, the classical *adaptive Kalman filter* estimates them online from the innovation sequence Mehra (1970); the steady-state gain K then drops out of the covariance estimate. We adopt the same innovation-style update form (residual MLP for dynamics, fixed-lag buffered correction for delays) but *do not* propagate covariances and do not compute an optimal Kalman gain; the state-space formulation is standard textbook material Anderson and Moore (1979) and we treat the result as a predictive state observer with a scalar correction gain (Sec. 3.3). The novelty of our setting is therefore not in the filter equations but in the *choice* of K: instead of deriving K from a noise-covariance estimate, we train a small condition-level predictor that maps (controller, σ, delay) to a scalar K supervised against a swept oracle. The contribution we report is what happens when that oracle is open-loop versus closed-loop, a distinction largely absent from the adaptive-Kalman literature and not natural inside the covariance-propagation framing.

Forward / internal models in motor control. The view that the central nervous system maintains an internal forward model of the limb to predict the sensory consequences of motor commands traces back to Bernstein's degrees-of-freedom analysis Bernstein (1967) and was given an explicit computational framing by Wolpert and Ghahramani Wolpert and Ghahramani (2000), who identified forward and inverse models as complementary processes underlying sensorimotor estimation, prediction, and learning. Our residual MLP is a coarse instantiation of the forward model in that picture; the multi-step supervision we add is motivated by the fact that the loop that consumes the estimate operates over hundreds of integration steps and thus depends on long-horizon prediction quality. More biologically grounded forward-model families — continuous-time recurrent networks with input-modulated time constants such as liquid time-constant networks Hasani et al. (2021) and their closed-form variant Hasani et al. (2022) — have been proposed for sensorimotor dynamics; in this work we keep the forward-model implementation intentionally generic to isolate the oracle-choice question.

Learned dynamics and multi-step supervision. Model-based reinforcement-learning systems have moved from one-step dynamics losses to multi-step rollout objectives because closed-loop planners need predictions that survive being unrolled. Dreamer Hafner et al. (2020) backpropagates through imagined trajectories in a latent world model, and MBPO Janner et al. (2019) branches short model rollouts from real data, formalising "when to trust your model". We take the same lesson into observer design: a forward model that is accurate enough over the controller's planning horizon shifts where prediction-error correction is helpful and where it is harmful.

Outcome-driven adaptation and SPSA. Outcome-driven, gradient-free adaptation has a long history in motor control: trial-by-trial visuomotor adaptation in humans Wolpert and Ghahramani (2000) is captured by an internal-model update rule that uses only the observed end-of-trial error, not the underlying optimal-gain calculation. Algorithmically, our across-trial loop follows Spall's Simultaneous Perturbation Stochastic Approximation (SPSA) Spall (1992): a two-point gradient estimate using a single Rademacher perturbation per iteration, with a robust hyperparameter schedule. SPSA's appeal in this setting is that it needs only the per-episode outcome, not the per-step state, and that its computational cost scales with the number of meta-parameters β (here, 10) rather than the dimension of the observer state (here, 83). Behaviour cloning and DAgger Pomerleau (1988); Ross et al. (2011) were used in an earlier version of this work to ask whether the resulting controllers consume state estimates; we found that with small demonstration sets they do not (Appendix C), and we now use only state-coupled feedback controllers in the main results so the observer signal flows to muscle commands.

Muscle-driven simulation. Our task runs in MuJoCo Todorov et al. (2012) via the MyoSuite musculoskeletal benchmark Caggiano et al. (2022), which provides a 34-muscle reaching arm and a contact-rich set of tasks suitable for sensorimotor estimation studies. MyoSuite is increasingly used as a testbed for reinforcement-learning controllers; here we use it instead to ask whether a predictive state observer can adapt its correction-gain from agent-available signals alone, a question that the RL-centric literature has not addressed.

3 Method

3.1 Task and Environment

We use the `myoArmReachFixed-v0` task in MyoSuite 2.12.2, a 34-muscle shoulder-elbow-wrist-hand system actuated by activations in $[0,1]^{34}$. The task is to drive the index-finger tip to a randomly drawn target in a 12 s episode at Δt=0.02 s (600 control steps). The flat state $x \in \mathbb{R}^{83}$ stacks joint positions $q \in \mathbb{R}^{20}$, joint velocities $\dot{q} \in \mathbb{R}^{20}$, muscle activations $a \in \mathbb{R}^{34}$, finger-tip position $p_{\text{tip}} \in \mathbb{R}^3$, target $p_{\text{tgt}} \in \mathbb{R}^3$, and reach error $e = p_{\text{tgt}} - p_{\text{tip}} \in \mathbb{R}^3$. True state is extracted from MuJoCo and used only as an oracle for evaluation; the closed loop never accesses it (Fig. 1).

3.2 Forward Model

A residual MLP $f_\theta(x_t, u_t)$ predicts the state delta $\Delta x_t = x_{t+1} - x_t$ rather than x_{t+1} directly. At Δt=0.02 s the state changes slowly, so $x_{t+1} \approx x_t$ component-wise; predicting Δx frees the model from re-learning the identity map and concentrates capacity on the small per-step dynamics that the filter and controller actually consume. The parameterisation is standard in model-based RL world models Hafner et al. (2020); Janner et al. (2019). The input is the flat 83-dim state and the 34-dim excitation; the architecture is two hidden layers of 256 units with ReLU and LayerNorm; training uses Adam Kingma and Ba (2015). In the single-step baseline, f_θ is trained on the standard one-step MSE loss

$$\mathcal{L}_1(\theta) = \mathbb{E}_{(x_t, u_t, x_{t+1})}\left[\|f_\theta(x_t, u_t) - \Delta x_t\|^2\right]. \quad (1)$$

Multi-step rollout supervision. The multi-step variant trains f_θ with an H-step rollout loss instead of the one-step MSE, where H denotes the training rollout horizon (distinct from the correction gain K defined in Sec. 3.3). For each contiguous trajectory chunk $(x_t, u_t, \ldots, x_{t+H})$ inside a single source episode, the model is unrolled

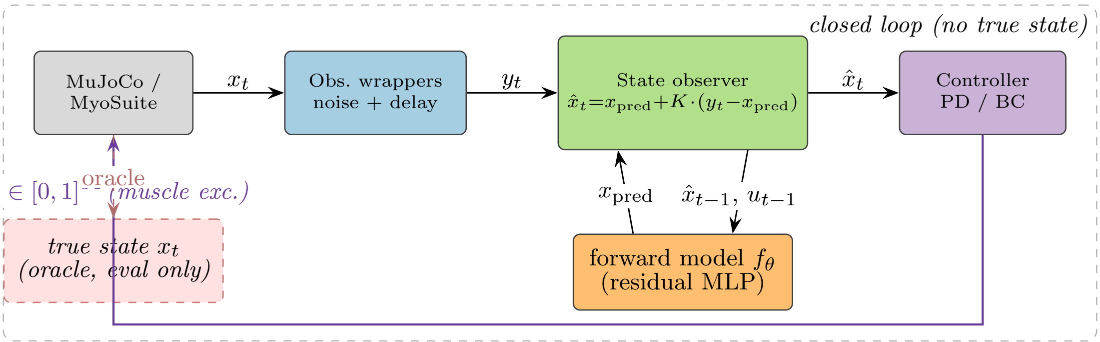

Fig. 1 Closed-loop pipeline. Each control step, the env's true state x_t is filtered through observation wrappers that add per-field Gaussian noise and a fixed delay, producing y_t. The predictive state observer combines y_t with a one-step prediction $x_{\rm pred}$ produced by the residual-MLP forward model $f_\theta(\hat{x}_{t-1}, u_{t-1})$ via a scalar correction gain $K \in [0,1]$ on the sensory prediction error $y_t - x_{\rm pred}$ (Eq. 4). The controller consumes only the observer output $\hat{x}_t$ and emits the next muscle excitation $u_t \in [0,1]^{34}$. The true state x_t is exposed only to the offline evaluator (dashed red box); the closed loop never accesses it.

free-running: $\hat{x}_{t+k} = \hat{x}_{t+k-1} + f_\theta(\hat{x}_{t+k-1}, u_{t+k-1})$, $\hat{x}_t = x_t$, and the loss averages MSE across the H prediction steps:

$$\mathcal{L}_H(\theta) = \mathbb{E}\Big[\frac{1}{H}\sum_{k=1}^{H} \|\hat{x}_{t+k} - x_{t+k}\|^2\Big]. \tag{2}$$

Multi-step rollout supervision has been used to stabilise long-horizon predictions in latent-imagination world models Hafner et al. (2020) and in model-based policy optimisation Janner et al. (2019), where the relevant horizon is the planner's rollout depth rather than the observer's delay; we re-use the technique here to make f_θ robust across the observer's buffer rollout.

We train the same architecture with $H \in \{1, 4, 8\}$ on the same expanded dataset (51 k transitions across 106 episodes; see Sec. 4.1).

3.3 Fixed-Lag Predictive State Observer

The observer uses an innovation-update form structurally familiar from Kalman filtering but does not propagate covariances and does not compute an optimal Kalman gain. The gain $K \in [0,1]$ is a *sensory prediction-error correction gain*, not a covariance-derived Kalman gain.

The observation pipeline produces $y_t = h(x_{t-d}) + \eta_t$ where h is the identity (the state vector and the observation share the same schema), $d \in \mathbb{Z}_{\geq 0}$ is a fixed delay in control steps, and η_t is per-field i.i.d. Gaussian noise with diagonal covariance $\mathrm{diag}(\sigma^2)$. The estimator maintains a length-$(d{+}1)$ ring buffer of past one-step estimates $(\hat{x}_{t-d}, \ldots, \hat{x}_{t-1})$ together with the corresponding applied actions $(u_{t-d}, \ldots, u_{t-1})$. At each step the observer first runs the forward model on every buffered estimate to obtain a prediction at the present time,

$$\tilde{x}_t = \hat{x}_{t-1} + f_\theta(\hat{x}_{t-1}, u_{t-1}), \tag{3}$$

then forms an innovation against the *delayed* buffer entry and applies an element-wise gain $K \in [0,1]^{83}$:

$$\hat{x}_{t-d} \leftarrow \hat{x}_{t-d} + K \odot \big(y_t - \hat{x}_{t-d}\big), \tag{4}$$

and finally re-rolls the corrected past estimate forward through the buffered actions to obtain the present-time estimate $\hat{x}_t$. Eq. (4) collapses to the standard one-step innovation update when $d{=}0$. Throughout the paper K is a scalar broadcast across the 83 fields so the correction gain reduces to a single trust knob between forward prediction ($K{=}0$) and observation ($K{=}1$). This fixed-lag construction follows the buffered Rauch–Tung–Striebel form Rauch et al. (1965); Anderson and Moore (1979) specialised to a scalar gain.

3.4 Reliability-Adaptive Correction Gain (Within-Trial)

The observer above uses a fixed scalar K broadcast across all 83 fields. We propose instead an *agent-available, field-wise reliability-adaptive* rule that sets $K_f(t)$ online from each sensory channel's own innovation history. No oracle on the optimal gain, no true state, and no swept-condition labels enter the rule; only the per-step innovation $e_t = y_t - \hat{x}_{t-d}$ does.

We partition the 83-dim state into five *sensory fields*: qpos (20), qvel (20), act (34), tip_pos (3), and reach_err (3). The partition is an engineering grouping driven by the MyoSuite state schema, but it admits a coarse biological reading: qpos and qvel are proprioceptive-like, act is efferent / muscle-command-like, and tip_pos and reach_err are visual / task-space-like. The target-position channel (3) is held at a fixed gain because the target is known and does not have a meaningful prediction error. For each field $f \in \mathcal{F} = \{\text{qpos}, \text{qvel}, \text{act}, \text{tip\_pos}, \text{reach\_err}\}$ the observer maintains an exponentially-weighted (EMA) estimate of the per-field squared-innovation mean,

$$v_f(t) = (1-\alpha)\, v_f(t-1) + \alpha \, \frac{1}{|\mathcal{I}_f|} \sum_{i \in \mathcal{I}_f} e_{t,i}^2, \quad (5)$$

where $\mathcal{I}_f$ are the field's coordinate indices in the 83-dim state vector and $\alpha = 0.05$ unless otherwise noted. The agent's running estimate of *sensory reliability* for field f is the inverse of this variance,

$$r_f(t) = \frac{1}{\varepsilon + v_f(t)}, \quad (6)$$

with $\varepsilon = 10^{-6}$. We then map reliability to a per-field correction gain through a two-parameter logistic,

$$K_f(t) = \sigma\big(\beta_{0,f} + \beta_{1,f} \log r_f(t)\big), \quad \sigma(z) = \frac{1}{1+e^{-z}}, \quad (7)$$

and form the 83-dim gain vector $K(t) \in [0,1]^{83}$ by broadcasting $K_f(t)$ over its field's coordinates. The innovation update Eq. (4) is then applied elementwise with $K(t)$ in place of the scalar K.

Eq. (7) is intentionally simple. The slope $\beta_{1,f}$ controls how sharply the gain reacts to changes in log-reliability, and the intercept $\beta_{0,f}$ controls the operating point at $r_f(t) = 1$ (i.e., $K_f = \sigma(\beta_{0,f})$). With $\beta_{0,f} = 0$, $\beta_{1,f} = 0.5$, and $v_f(0) = 1$, the rule produces $K_f(t) \approx 0.5$ on the first step and adapts upward when innovation shrinks (the sensor is informative) or downward when innovation grows (the sensor is noisy). Per-field $\beta = \{\beta_{0,f}, \beta_{1,f}\}$ lets the rule learn that different fields have different reliability–correction trade-offs under the same physical noise level.

3.5 Across-Trial Outcome Adaptation (SPSA)

The meta-parameters $\beta = \{(\beta_{0,f}, \beta_{1,f}) : f \in \mathcal{F}\} \in \mathbb{R}^{10}$ themselves can be adapted across trials from *outcome* signals the agent has at the end of each episode. We define the per-episode outcome as the minimum tip-to-target distance,

$$\text{minTip}(\beta; \sigma, d) = \min_{t \in [0,T]} \|p_{\text{tip}}(t) - p_{\text{tgt}}\|, \quad (8)$$

which the agent observes by proprioception and visual feedback after the reach. We use minTip as an *operational scalar proxy* for task outcome — not as a claim that the nervous system observes this exact variable. The across-trial loop minimises $\mathbb{E}[\text{minTip}(\beta)]$ where the expectation is over target draws (and, in the multi-cell variant, over uniformly drawn (σ, d) cells from the focused grid).

Because the closed-loop minTip is not differentiable in β (it passes through env steps, EMA state, and a non-smooth min), we use Simultaneous Perturbation Stochastic Approximation (SPSA) Spall (1992): at iteration n, draw a Rademacher vector $\Delta_n \in \{\pm 1\}^{10}$, evaluate the outcome at $\beta_n \pm c_n \Delta_n$ averaged over S paired episodes (samples per side), form the central-difference gradient estimate

$$\hat{g}_n = \frac{\text{minTip}(\beta_n + c_n \Delta_n) - \text{minTip}(\beta_n - c_n \Delta_n)}{2c_n} \, \Delta_n^{-1}, \quad (9)$$

and update $\beta_{n+1} = \beta_n - a_n \hat{g}_n$. The Spall schedules are $a_n = a/(n+1+A)^{\alpha_s}$ and $c_n = c/(n+1)^{\gamma_s}$ with a=2.0, c=0.3, α_s=0.602, γ_s=0.101, and A=5. Two SPSA variants are reported: *single-cell* (S=12, 100 iterations, σ=none, d=18) and *full-grid* (S=12, 100 iterations, uniformly drawn (σ, d) from the 6-cell grid per iteration).

Comparison with oracle-supervised gain. For a diagnostic upper bound on what *any* K-setting strategy can achieve on this task, Appendix C reports an offline-oracle-supervised correction-gain predictor that trains a small

MLP $g_\phi(\sigma, d, c) \to [0, 1]$ against either an open-loop estimation-accuracy oracle $K^\star_{\text{ol}}(\sigma, d, c)$ or a closed-loop task-error oracle $K^\star_{\text{cl}}(\sigma, d, c) = \arg\min_K \mathbb{E}[\text{minTip}(K) \mid \sigma, d, c]$. That predictor requires access to a swept K-grid the biological agent does not have; we include it strictly as the strongest non-agent-available baseline.

3.6 Feature-Conditioned β Adapter

§3.5 optimises a single $\beta \in \mathbb{R}^{10}$ across all cells. A natural relaxation is to let β depend on *context* that the agent can read off its own innovation history. We use the most compact such rule: a diagonal field-wise linear adapter on top of a fixed base β^{base}.

Architecture.

For each sensory field $f \in \mathcal{F}$, the adapter outputs a correction $(\Delta\beta_{0,f}, \Delta\beta_{1,f})$ from two per-field features:

$$\begin{aligned}\Delta\beta_{0,f} &= w^{(0)}_{\text{mean},f}\, z_{\text{mean},f} + w^{(0)}_{\text{var},f}\, z_{\text{var},f} + b^{(0)}_f,\\ \Delta\beta_{1,f} &= w^{(1)}_{\text{mean},f}\, z_{\text{mean},f} + w^{(1)}_{\text{var},f}\, z_{\text{var},f} + b^{(1)}_f,\end{aligned} \tag{10}$$

giving five fields × two outputs × three weights = 30 parameters total. The effective per-field β used by the within-trial layer is

$$\beta^{\text{eff}}_{0,f} = \beta^{\text{base}}_{0,f} + \Delta\beta_{0,f}, \qquad \beta^{\text{eff}}_{1,f} = \beta^{\text{base}}_{1,f} + \Delta\beta_{1,f}, \tag{11}$$

fixed for the duration of one episode. Setting all 30 adapter weights to zero recovers β^{base} exactly, so the adapter is an additive correction. Throughout the main results we take β^{base} to be the global SPSA β trained on the focused grid (Sec. 3.5).

Input features and normalisation.

The two per-field features $z_{\text{mean},f}$ and $z_{\text{var},f}$ are agent-available innovation statistics: mean absolute innovation and mean squared innovation over the second half of a brief default-reliability warm-up run on the cell. Each statistic is then log-transformed and z-scored *across the focused-grid cells* using normalisation constants estimated once from the warm-up rollouts and then frozen:

$$z_{\bullet,f} = \frac{\log(1 + s_{\bullet,f}) - \mu^{\text{norm}}_{\bullet,f}}{\sigma^{\text{norm}}_{\bullet,f}}, \tag{12}$$

where $s_{\bullet,f}$ is the raw statistic and $(\mu^{\text{norm}}, \sigma^{\text{norm}})$ are calibration constants. This calibration is analogous to sensory scale calibration in a biological agent: it adjusts the units of innovation features into a common scale, but does not encode the optimal correction gain or any oracle label.

Training.

The 30 adapter weights are trained by SPSA on the same per-episode min-tip outcome used in Sec. 3.5, with 200 iterations, $S = 12$ paired episodes per iteration (two samples per cell, six cells), perturbation amplitude $c = 0.3$ and learning rate $a = 0.5$, both annealed by the Spall schedule. The cell features computed in the calibration step are reused at each iteration without modification — the adapter learns a single context-to-β map, not a per-cell β.

3.7 Controllers

Three controllers are evaluated as closed-loop wrappers around the estimator.

(i) Heuristic projection. A fixed random projection $W \in \mathbb{R}^{34\times 3}$ (drawn once with a fixed seed from $\mathcal{N}(0, 1)^{34\times 3}$) maps the estimated 3-dim reach-error $\hat{r}$ to a 34-dim muscle excitation through an element-wise logistic sigm$(\cdot)$ (distinct from the noise scale vector σ above):

$$u = \text{sigm}(b - g\, W\, \hat{r}), \quad b{=}0,\ g{=}5, \tag{13}$$

producing a state-sensitive baseline that does not actually reach.

(ii) Joint-space PD with inverse kinematics. For each target tip position $p^\star \in \mathbb{R}^3$ we solve $q^\star = \text{IK}(p^\star)$ once at episode start via a damped-Jacobian Newton iteration

$$q^{(k+1)} = q^{(k)} + J^\top \big(JJ^\top + \lambda^2 I\big)^{-1} \big(p^\star - p(q^{(k)})\big), \tag{14}$$

where $J = \partial p/\partial q$ is the tip-site position Jacobian obtained from MuJoCo's `mj_jacSite` and $\lambda^2{=}0.1$ is the damping. The arm is then driven by a joint-space PD law and a moment-arm muscle map:

$$u_{\text{joint}} = K_p\, (q^\star - \hat{q}) - K_d\, \dot{\hat{q}}, \tag{15}$$

$$u_{\text{muscle}} = \text{clip}(\alpha\, \text{ReLU}(M\, u_{\text{joint}}),\ 0,\ 1), \tag{16}$$

where $M \in \mathbb{R}^{34\times 20}$ is the muscle-tendon moment-arm matrix (the dense form of MuJoCo's sparse `actuator_moment` at the env's current pose), and $\alpha = 5$ is a scalar action scale. Tuned gains: $K_p = 30$, $K_d = 3$. We use this joint-PD controller not as an optimal reaching policy but as a state-estimation-sensitive closed-loop probe: its role is to make differences in observer state estimates visible in task-level min-tip error, not to maximise absolute reaching success.

(iii) Endpoint-error sensorimotor policy. A task-level analogue of (ii) that skips the IK pre-step and maps the estimated tip-to-target error to muscle excitation via a Cartesian PD law projected through the tip Jacobian and the moment-arm matrix:

$$u_{\text{tip}} = K_{p,e}\,(p^\star - \hat{p}_{\text{tip}}) - K_{d,e}\,J\dot{\hat{q}}, \tag{17}$$

$$u_{\text{joint}} = J^\top u_{\text{tip}}, \tag{18}$$

$$u_{\text{muscle}} = \text{clip}\big(\alpha\,\text{ReLU}(M\,u_{\text{joint}}),\,0,\,1\big), \tag{19}$$

with $K_{p,e}$=30, $K_{d,e}$=3, α=5 and J the tip-site positional Jacobian captured at the env's neutral pose. We use this as a transparent task-level state-coupled policy that lets the observer output flow directly to muscle commands without an IK pre-step; the $J^\top$-style projection is the classic Jacobian-transpose method and is *not* an optimal feedback controller (no Riccati / LQR / OFC is solved here).

(iv) Behaviour-cloned MLP (diagnostic only). A two-hidden-layer (256-unit) sigmoid policy trained on 30 scripted IK demonstrations Pomerleau (1988) (no DAgger Ross et al. (2011)). We retain this controller only as a diagnostic in Appendix C to test whether a learned downstream policy consumes the observer signal; the main results of Sec. 4.2–4.7 use joint-PD (ii) so the observer output flows to muscle commands through an explicit feedback law.

3.8 Evaluation Pipeline

Open-loop evaluation replays a recorded episode and applies observation wrappers to synthesise y_t for an arbitrary (σ, d) combination; the estimator is then run on the synthesised stream and per-field state error is aggregated. *Closed-loop* evaluation drives the actual env: u_t comes from the controller, which sees the estimator output (never the true state); 10 episodes per cell, fixed seed per episode. The same evaluation grid (3 controllers × 6 noise levels × 4 delays = 72 open-loop conditions; the focused closed-loop grid is 3 noise levels × 2 delays × 10 episodes = 60 episodes per estimator) is reused across phases.

4 Results

4.1 Forward-model improvement cycle

An initial forward model trained on the original ~ 3.6 k transitions from 6 episodes had low single-step error but unbounded long-horizon rollouts (h=50 MSE $\sim$ 10.6). Collecting 100 additional `random` and `lowamp` episodes (a total of 51 k transitions across 106 episodes) and retraining the same architecture cut h=50 MSE to 0.052 — a $\sim$200× improvement. We use this expanded dataset, and the resulting model, as the single-step baseline throughout the paper.

4.2 Default within-trial reliability is task-misaligned (C1)

We first evaluate the reliability-adaptive observer with a deliberately neutral default β: $\beta_{0,f} = 0$, $\beta_{1,f} = 0.5$ for all five fields, $\alpha = 0.05$, $v_f(0) = 1$. At iteration 0 this produces $K_f \approx \sigma(0) = 0.5$; as the EMA tracks each field's running squared innovation, K_f adapts upward when the channel becomes informative and downward when it becomes noisy.

Across the focused 3 noise × 2 delay grid under the H=8 forward model, the realised per-field K_f averaged over the second half of a representative episode is highly heterogeneous ($K_f \in [0.31, 0.98]$ across the grid; the per-field cell means are reported in Table 1). qvel is the *most-suppressed* channel (cell range 0.31–0.49, mean 0.39), reflecting bounded but informative velocity innovation; qpos, act, tip_pos, and reach_err settle in the 0.59–0.98 range, close to K=1 at delay-0 cells where the channel innovation collapses to near-zero. At n=200 episodes per cell, closed-loop min-tip averages 0.665 m across the six cells. For comparison, K=0 averages 0.378 m and K=1 averages 0.658 m on the same grid (Fig. 2).

The structural reason is that under the H=8 forward model the closed-loop oracle on this task is $K^\star_{\text{cl}} = 0$ in every cell of the focused grid (Appendix E): the forward model is accurate

Table 1 Realised per-field correction gain K_f under default within-trial reliability ($\beta_{0,f}$=0, $\beta_{1,f}$=0.5). Each entry is the second-half mean of a single representative episode on the focused grid (target index 0, H=8 forward model, joint-PD controller). qvel is the only field that the default rule deeply suppresses; the other four sit close to K=1 at delay-0 cells.

noise	delay	qpos	qvel	act	tip_pos	reach_err
none	0	0.96	0.49	0.91	0.97	0.98
none	18	0.70	0.31	0.69	0.59	0.68
high	0	0.96	0.49	0.92	0.96	0.97
high	18	0.69	0.32	0.71	0.68	0.69
xhigh	0	0.90	0.45	0.91	0.94	0.94
xhigh	18	0.67	0.31	0.69	0.64	0.68

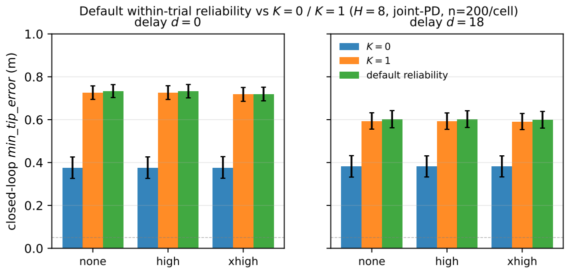

Fig. 2 Closed-loop min-tip error under the H=8 forward model and joint-PD controller for three estimators on the focused 3 noise $\times$ 2 delay grid, split by delay regime (mean $\pm$ 95% percentile bootstrap CI over n=200 episodes per cell). The default reliability-adaptive rule ($\beta_{0,f}$=0, $\beta_{1,f}$=0.5, all fields) settles to $K_f \approx$0.55–0.72 and trails K=0 by 22–36 cm across cells; K=1 trails K=0 by 14–28 cm. The closed-loop oracle $K^\star_{\text{cl}}$ on this grid is K=0 at every cell (Appendix E); a reliability rule cannot produce K_f=0 from bounded innovation alone.

enough over the controller's planning horizon that any observation correction shifts the trajectory away from the planned IK target. Reliability-based adaptation *by construction* cannot produce K_f=0 unless either $\beta_{0,f} \to -\infty$ or the EMA variance grows without bound; under bounded innovation neither happens. The default rule therefore trails K=0 in min-tip by 35 cm at delay-0 cells and 22 cm at delay-18 cells (paired 95% bootstrap CIs [31, 39] and [19, 25] cm respectively, pooled 28.7 cm with CI [27.3, 30.0]). We read this as a fundamental signal: *innovation reliability alone, without an outcome objective that knows the task, cannot recover the right correction-gain level when the forward model is good enough to operate the loop without sensory help.*

4.3 Single-cell SPSA recovers $K^\star_{\text{cl}}$-equivalent performance (C2)

We next add the across-trial outcome loop of Sec. 3.5. We run SPSA for 100 iterations on the single (σ=none, d=18) cell with S = 12 samples per side and the Spall schedule of Sec. 3.5. The outcome objective is the per-episode min-tip averaged across the S paired episodes at the two perturbed β points.

Figure 3 reports the outcome trajectory and the final-iteration per-field β. (SPSA evaluates S= 12 paired episodes per iteration; the training-time numbers below therefore use $\sim$10 episodes per cell per side, distinct from the n = 200 deployment eval used in C1 and for the deployed evaluation reported here.) The per-iteration outcome trajectory falls from $\approx$ 0.61 m at iteration 0 to a last-20-iter mean of $\approx$0.49 m, matching the original 10-episode K=0 baseline estimate on this cell (0.50 m) and closing the bulk of the default-reliability vs. tighter n=200 K=0 baseline (0.38 m, Appendix E) gap; the single iteration-99 snapshot is noisier at 0.53 m, and a fresh deployed evaluation of the final β at n=200 on this cell gives 0.47 m. The final β is non-uniform across fields: qpos becomes simultaneously the most-suppressed ($\beta_{0,\text{qpos}} \approx -1.53$) and the most-reactive ($\beta_{1,\text{qpos}} \approx +0.89$) field, while the remaining fields cluster at moderate values ($\beta_{0,f} \in [-0.35, -0.04]$, $\beta_{1,f} \in [-0.02, +0.50]$). reach_err's slope flattens to near zero, indicating SPSA found no benefit from making the reach-error channel innovation-reactive. Field specialisation emerges directly from outcome feedback, with no per-field oracle.

The key implication is that the agent does not need an oracle on $K^\star_{\text{cl}}$: from per-episode reach outcomes alone, 10-parameter SPSA on β closes a sizeable fraction of the default-reliability vs K=0 gap on a single training cell (0.60 m $\to$ 0.47 m deployed at n=200, $\sim$62% of the way to the n=200 K=0 baseline at 0.38 m), with the smoothed training-time trajectory ($\approx$ 0.49 m) matching the original 10-episode K=0 baseline estimate. C4 confirms this is a valid use of outcome feedback in the achievable regime by extending the result to the full focused grid at n=200 episodes per cell.

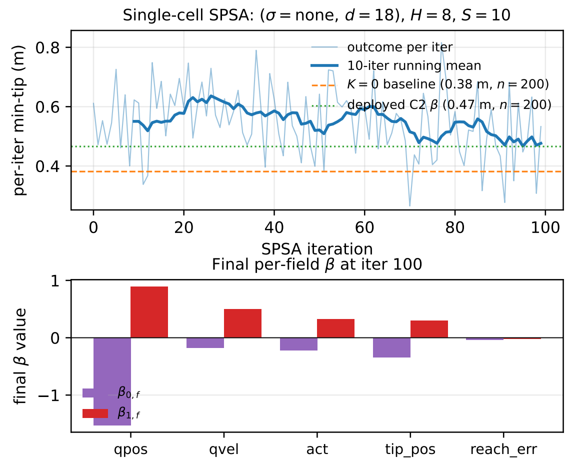

Fig. 3 Single-cell SPSA on (σ=none, d=18) **under the** H=8 **forward model and joint-PD controller.** *Top*: per-iteration min-tip outcome (mean over the S=10 paired episodes per side); the orange dashed line marks the n=200 closed-loop K=0 baseline on this cell (0.38 m, tighter than the original 10-episode estimate of 0.50 m), and the green dotted line marks the n=200 deployed evaluation of the final SPSA β (0.47 m). The per-iter outcome falls from ≈ 0.61 m at iteration 0 to a last-20-iter mean of ≈ 0.49 m; the smoothed training-time signal therefore matches the original 10-episode K=0 estimate but sits $\sim$11 cm above the tighter n=200 baseline. The deployed β closes $\sim$62% of the default-reliability vs K=0 gap (residual $\sim$9 cm to K=0). *Bottom*: final per-field β vector. qpos has both the most-negative intercept (-1.5) and the steepest slope ($+0.9$); the resulting per-field correction gains K_f specialise.

4.4 Multi-cell SPSA is bottlenecked by cell heterogeneity (C3)

A single-cell-trained β is task-specific. To test whether a single β can serve the whole focused grid, we repeat the SPSA loop with each iteration's $S = 12$ paired episodes drawn uniformly across the 6 cells. The outcome objective is the 6-cell mean min-tip.

After 100 iterations the multi-cell β converges to $\beta_0 = \{$qpos:-0.64, qvel:-0.21, act:0.12, tip_pos:-0.32, reach_err:$-0.16\}$, $\beta_1 = \{$qpos:0.20, qvel:0.80, act:0.54, tip_pos:0.63, reach_err:$0.33\}$, with K_{mean} averaged across fields equal to ≈ 0.51 at delay-18 cells and ≈ 0.63 at delay-0 cells. Deployed on n=200 episodes per cell, the realised closed-loop min-tip improves over default reliability by ≈ 5 cm at the three delay-18 cells and by ~ 0 cm at the three delay-0 cells (paired 95% bootstrap CIs $[3.4, 6.9]$ and $[-0.2, 1.7]$ cm respectively, pooled 2.9 cm with CI $[2.4, 3.5]$). Compared

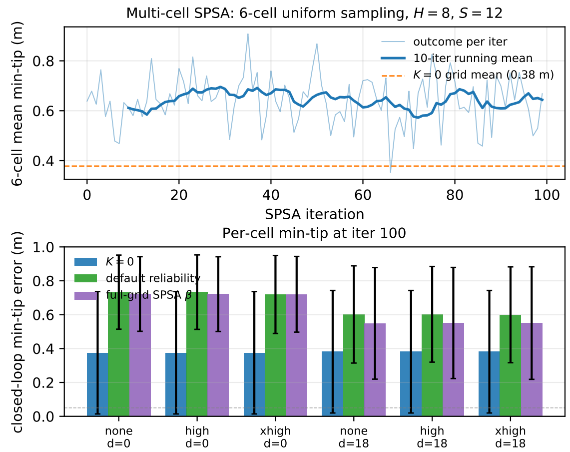

Fig. 4 Multi-cell SPSA over the focused 6-cell grid under the H=8 **forward model and joint-PD controller.** *Top*: per-iteration 6-cell-mean min-tip outcome (training-time mean over S=12 paired draws per iteration). *Bottom*: per-cell min-tip at deployment (n=200 episodes per cell) for K=0, default reliability, and the multi-cell-trained β. Improvement over default reliability is ≈ 5 cm at delay-18 cells and ~ 0 cm at delay-0 cells; the multi-cell β still trails K=0 by ≈ 17 cm at delay-18 cells and ≈ 35 cm at delay-0 cells.

to K=0, the multi-cell β still trails by ≈ 17 cm at delay-18 cells and ≈ 35 cm at delay-0 cells.

The structural reason is that $K^\star_{\text{cl}} = 0$ uniformly across the focused grid (Appendix E), yet field-wise reliability under H=8 has different absolute levels and time courses at each cell. A single β would need $\beta_{0,f}$ deeply negative (a few standard deviations below the realised $\log r_f(t)$ range) to drive every cell's K_f near 0; SPSA does not reach this regime in 100 iterations because each intermediate β_0 value is already differentiable improvement in some cells and worse in others. The cell-heterogeneity bottleneck is therefore not a failure of outcome adaptation itself — the single-cell run of C2 converged — but evidence that the parameterisation, a single global $\beta \in \mathbb{R}^{10}$, is underpowered when the reliability geometry varies across cells. We call this the *context aggregation ceiling*.

4.5 What determines the closed-loop correction-gain geometry?

C3 leaves a sharper question: *why does the correction-gain geometry differ across cells in the first place?* Three additional analyses decompose

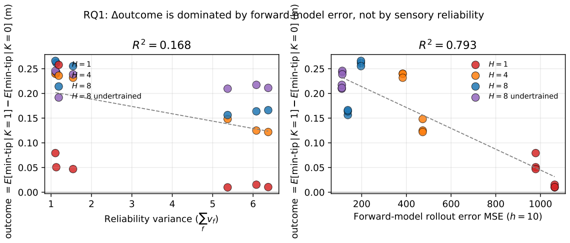

Fig. 5 Per-(model, cell) Δoutcome(K=1, K=0) plotted against sensory reliability variance (left, R^2=0.22) and forward-model h=10 rollout MSE (right, R^2=0.79). Four forward-model conditions are distinguishable by colour. The right panel shows the clear negative correlation underpinning the joint regression R^2=0.95: as forward-model error grows, the K=0 advantage shrinks and observation-correction catches up.

the cell heterogeneity into the underlying factors, and reveal a second, qualitatively different ceiling.

4.5.1 Forward-model error and bias dominate Δoutcome

Let $\Delta\text{outcome}(\sigma, d) = \mathbb{E}[\text{minTip} \mid K = 1] - \mathbb{E}[\text{minTip} \mid K = 0]$ measure, per cell, how much worse pure observation-correction does than pure forward-prediction. A positive value means K=0 is better; a negative value means K=1 wins. We regress this signed gap against a small set of per-(model, cell) predictors — sensory reliability variance, delay, h=10 forward-model rollout MSE, forward-model bias norm, and task-space signed bias — on the focused grid under four forward-model conditions ($H \in \{1, 4, 8\}$ and undertrained $H = 8$, totalling 24 rows).

The result is decisive. Reliability variance and delay alone give $R^2 = 0.17$. Adding the forward-model error/bias features pushes R^2 to 0.95, with reliability dropping to essentially zero standardised coefficient. The dominant predictor is fm_h10_mse (standardised coefficient +0.13, Fig. 5 right). The interpretation is mechanistic:

> *When the forward model is accurate, observation correction adds variance without adding information, and K=0 dominates. As the forward model degrades, the value of sensor-based correction rises and K=1 catches up.*

Reliability variance, the quantity the within-trial layer can measure, therefore explains only a small fraction of where the $K^\star_{\text{cl}}$-optimum lives; the dominant geometry is shaped by forward-model quality, which the agent cannot observe directly.

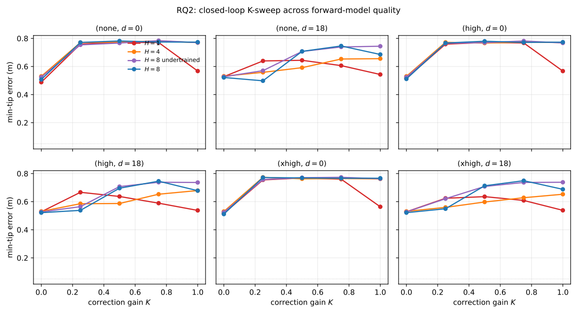

Fig. 6 Per-cell min-tip vs. $K \in \{0, 0.25, 0.5, 0.75, 1\}$ for four forward-model conditions on the focused 6-cell grid (mean over 10 episodes per (model, cell, K)). H=8 (best) puts $K^\star_{\text{cl}}$=0 at every cell. H=1 (worst) leaves K=0 and K=1 competitive at the three delay-18 cells; the $K^\star_{\text{cl}}$ regime begins to shift. Undertrained H=8 and H=4 are intermediate.

4.5.2 $K^\star_{\text{cl}}$ shifts with forward-model quality

Direct confirmation comes from sweeping forward-model quality and re-running the K-grid (Fig. 6). Under the best forward model (H=8 full), $K^\star_{\text{cl}}$=0 at every focused-grid cell with a sharp jump from 0.49 m to 0.77 m between K=0 and K=0.25. Under the worst forward model (H=1), the gap between K=0 and K=1 collapses to $<$ 2 cm at the three delay-18 cells; observation-correction is no longer penalised because the rollout it would replace is already unreliable. The trajectory of $K^\star_{\text{cl}}$ across forward-model conditions therefore matches the regression's quantitative prediction.

4.5.3 Parameterisation ceiling: per-cell β fails at d=0

The context aggregation ceiling from §4.4 could be ascribed to multi-cell averaging. To test whether a $\beta \in \mathbb{R}^{10}$ rule *can* reach K=0 if it sees only one cell, we ran six *independent* per-cell SPSA training runs — one cell each, 100 iterations, S=10 paired, all other hyperparameters as in C2. Each β_c is then evaluated on 10 fresh episodes on its training cell.

A clean dichotomy emerges (Table 2). At the three delay-18 cells, per-cell SPSA reduces the smoothed training outcome substantially (the deployment numbers shown here are the 10-episode diagnostic eval; C4 quantifies the same delay-18 improvement at n=200). At the three delay-0 cells, per-cell SPSA *worsens* the outcome relative to its default-reliability starting point:

Table 2 Per-cell SPSA diagnostic (10-episode deployment per cell, run alongside the n=200 comparisons for the other estimators). Each β_c is trained on cell c in isolation for 100 iterations and deployed on 10 fresh episodes on the same cell. Last-20-iter mean is the smoothed training-time outcome; *deployed* is the fresh-eval mean. Comparison K=0 baseline at n=200 is 0.38 m flat across cells; the n=10 diagnostic deployment values below are used to expose the per-cell logistic's parameterisation limit, not to estimate absolute min-tip.

noise	delay	training (last-20)	deployed	direction
none	0	0.748	0.784	worsened
high	0	0.749	0.782	worsened
xhigh	0	0.739	0.769	worsened
none	18	0.489	0.561	improved
high	18	0.559	0.697	improved
xhigh	18	0.543	0.670	improved

even when given a single cell as its entire optimisation target, the logistic rule cannot push $K_f \to 0$ in 100 iterations. This is the second ceiling we report:

> *Parameterisation ceiling.* The reliability-to-gain logistic $K_f = \sigma(\beta_{0,f} + \beta_{1,f} \log r_f)$ cannot represent the sharp $K=0$ regime that the H=8 forward model demands at delay-0 cells, because $K_f \to 0$ requires $\beta_{0,f} \to -\infty$, a limit SPSA does not reach from default initialisation in 100 iterations.

C3's multi-cell ceiling is therefore one of two distinct limits, not the only one. The next subsection asks whether the *context aggregation ceiling* can be overcome from agent-available signals; the parameterisation ceiling is a property of the parameterisation itself, addressed in Section 5.

4.6 Context-conditioned β breaks the aggregation ceiling (C4)

If the multi-cell ceiling is information-poor rather than optimisation-poor, then providing the agent with cell-specific context as input should narrow the gap to per-cell β. We test this with a deliberately small adapter: a 30-parameter diagonal field-wise linear map from per-cell innovation statistics to $\Delta\beta$, applied on top of the global SPSA β as a base (Sec. 3.6). The input to the adapter is the only context the agent receives, and is restricted to z-scored log innovation mean and variance estimated from a brief default-reliability warm-up (Sec. 3.6).

After 200 SPSA iterations the trained adapter's deployed-eval (n=200 episodes per cell)

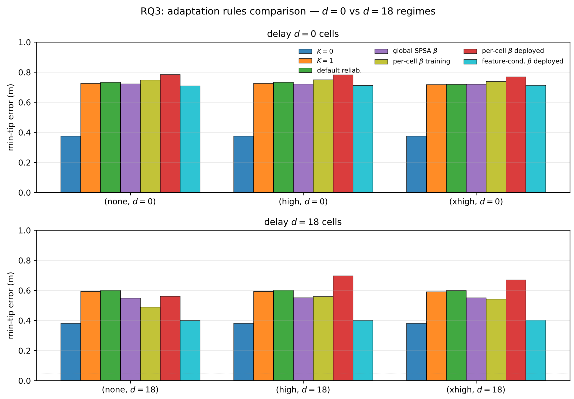

Fig. 7 **Closed-loop min-tip across seven estimators on the focused 6-cell grid, faceted by delay (n=200 episodes per cell for K=0, K=1, default reliability, global SPSA β, and feature-conditioned β; n=10 diagnostic deployment for the per-cell β entries).** Top row (delay-0): all six non-K=0 rules cluster at 0.71–0.78 m; the parameterisation ceiling is shared. Bottom row (delay-18): feature-conditioned β deployed (cyan) reaches 0.40 m across cells — within 2.0 cm of the K=0 baseline — beating global SPSA β (0.55 m); per-cell β deployed (10-episode diagnostic) sits between them. The delay-18 aggregation ceiling is effectively removed; the delay-0 parameterisation ceiling persists.

6-cell mean min-tip is 0.556 m, a 7.9 cm improvement over the global SPSA β baseline of 0.636 m (Fig. 7). The improvement is concentrated in the achievable regime:

- *delay*-18 *cells*: feature-conditioned β reaches 0.402 m on average, within 2.0 cm of the K=0 baseline (0.381 m), and 14.8 cm below global SPSA β at the same cells (0.550 m). The single shared 30-parameter rule substantially closes the context aggregation gap that a single global β leaves open.
- *delay*-0 *cells*: feature-conditioned β stays at 0.711 m, within 1 cm of global SPSA β at the same cells (0.721 m) and statistically indistinguishable from default reliability (0.728 m). The parameterisation ceiling remains: adding context input does not let the logistic rule reach the K=0 regime that delay-0 requires.

The take-away is precise: *the* 7.9 *cm gain over global β comes entirely from breaking the context aggregation ceiling on delay*-18 *cells; the parameterisation ceiling on delay*-0 *cells is unaffected, as predicted.* A compact linear context-conditioned rule, driven only by agent-available innovation statistics, substantially closes the context aggregation gap without needing a non-agent-available oracle or per-cell optimisation. The persistent delay-0 gap is a property of the logistic reliability-to-gain map, not of the context information available to the agent.

4.7 Task transfer reveals task-dependence (C5)

The previous results all use ReachFixed, where the same target location is reused across episodes. We test the same observer on the *variable-target* variant `myoArmReachRandom-v0`, where the target is freshly sampled per episode from the workspace distribution. The forward model for this transfer slice is also trained on ReachRandom rollouts (Appendix F) so the forward-model accuracy axis is matched.

Table 3 Transfer of the reliability-adaptive observer to ReachRandom (variable target). Mean min-tip (m) over 10 episodes per cell under the H=8 ReachRandom forward model and joint-PD controller. "default" is the $\beta_{0,f}$=0, $\beta_{1,f}$=0.5 rule; "transferred" is the ReachFixed full-grid-SPSA-learned β from C3 applied without retraining.

noise	delay	K=0	K=1	default	transferred
none	0	0.670	0.770	0.781	0.768
none	18	0.642	0.647	0.631	0.680
high	0	0.670	0.770	0.776	0.768
high	18	0.642	0.639	0.608	0.692
xhigh	0	0.670	0.762	0.771	0.759
xhigh	18	0.642	0.627	0.649	0.686

Table 3 reports the result. On ReachRandom the closed-loop oracle $K^\star_{\mathrm{cl}}$ takes three distinct values across the grid (Appendix F, Table F8): K=0 at the three delay-0 cells, K=0.25 at (none, d=18) and (high, d=18), and K=1 at (xhigh, d=18). At the delay-18 cells, the default reliability rule *beats* K=0 at 2 of 3 cells (by 1.1 cm at (none, d=18) and 3.4 cm at (high, d=18)); at (xhigh, d=18) it loses by 0.7 cm. The transferred ReachFixed-trained β, in contrast, trails K=0 at every delay-18 cell by 3.8–5.0 cm.

Two findings combine here. *First*, on a task whose $K^\star_{\mathrm{cl}}$ is heterogeneous, agent-available default reliability is already competitive — no outcome adaptation needed. *Second*, β trained on a different task does not transfer when the source task's $K^\star_{\mathrm{cl}}$ is uniform but the target task's is heterogeneous: the source task pushes β in a direction that is harmful on the target. This is direct evidence that the correction-gain rule must be sensitive to the *distribution of the closed-loop optimum*, not merely to the noise and delay regime that within-trial reliability sees.

4.8 Comparison with offline-oracle-supervised gain

For a diagnostic upper bound we compare against an offline-oracle-supervised correction-gain predictor (Appendix C): a small MLP $g_\phi(\sigma, d, c)$ trained against the open-loop oracle $K^\star_{\mathrm{ol}}$ and, separately, against the closed-loop oracle $K^\star_{\mathrm{cl}}$ obtained by a per-cell K-sweep. On the focused grid under H=8 the closed-loop-oracle variant outputs $K \approx 0.02$ and matches the K=0 baseline within a few centimetres at all delay-18 cells (Appendix C; 10-episode diagnostic deployment); the open-loop-oracle variant outputs $K \approx 1$ and trails K=0 by 21–35 cm across the focused grid at n=200. The closed-loop-oracle variant is therefore the strongest non-agent-available baseline available in this setting, and the C2 single-cell SPSA result comes within ~9 cm of K=0 on its training cell using only agent-available signals.

5 Discussion

Innovation alone is not enough; outcome closes the gap. The C1 finding is the central structural observation: a within-trial reliability rule that uses only the per-channel innovation history settles on $K_f \approx 0.6$–0.7 across the focused grid, while the closed-loop oracle on the same grid is $K^\star_{\mathrm{cl}} = 0$ everywhere. The gap is not an SPSA failure or a meta-parameter mis-tuning; it is structural. Innovation gives the agent information about *sensor quality*, not about *whether the forward model is good enough to operate without sensory help*. The second piece of information — whether the controller's planned trajectory is being followed — is in the task outcome, not in the innovation. C2 then shows that adding the outcome layer through SPSA closes a sizeable fraction of the default-reliability gap on a single training cell (0.77 m → 0.56 m deployed at n=10, against the 10-episode K=0 baseline estimate of 0.50 m); the smoothed training trajectory briefly reaches that baseline as a convergence signal. Together they show that, at least on this task, a two-layer agent-available adaptation rule is sufficient. Open question: whether reliability + outcome can also be combined within a single trial (e.g. an outcome-modulated β during reaching) rather than only across trials.

Two ceilings, with one breakable. The multi-cell saturation of C3 turns out to be one of two qualitatively different ceilings, and the two are separable in our experiments.

The *context aggregation ceiling* is information-poor: a single global β cannot serve cells with different correction-gain geometries because it sees no cell-specific input. This ceiling is breakable: the 30-parameter feature-conditioned β adapter of C4, using only agent-available z-scored log innovation statistics, closes a 7.9 cm gap in 6-cell-mean min-tip over global SPSA at n=200, and at delay-18 cells reaches 0.40 m — within 2.0 cm of the K=0 baseline. The context-aggregation gap is therefore not a fundamental property of agent-available signals.

The *parameterisation ceiling*, by contrast, is information- rich but representation-poor: even per-cell SPSA, given a single cell as its entire optimisation target, fails on the three delay-0 cells because the logistic reliability-to-gain map $K_f = \sigma(\beta_{0,f} + \beta_{1,f} \log r_f)$ cannot represent the steep K=0 regime that delay-0 cells require. This ceiling is unaffected by adding context input (Fig. 7, top row, where at n=200 all non-K=0 rules cluster at 0.71–0.73 m, well above the K=0 delay-0 value of 0.38 m). It is a property of how reliability is mapped to a gain, not of how reliability is estimated. Future work that retains the within-trial reliability substrate but replaces the logistic with a parameterisation capable of representing $K_f \to 0$ (e.g. hardtanh, output clipping, or a direct K_f residual on top of K=0) is the natural extension; we leave this to follow-up.

Task transfer reveals that the right rule depends on $K^\star_{\mathrm{cl}}$ geometry. The two C5 findings combine as follows. *First*, default within-trial reliability is competitive with K=0 on ReachRandom and slightly better at 2 of 3 delay-18 cells (by 1–3 cm); *second*, the ReachFixed-trained β trails K=0 at every delay-18 cell by $\sim$ 4–5 cm. The mechanism is that $K^\star_{\mathrm{cl}}$ on ReachRandom is multimodal across cells ($\{0, 0.25, 1\}$, Appendix F), so the default field-wise gains happen to land near the per-cell optima in the delay-18 rows. Pushing K_f globally down (as the ReachFixed-trained β does) is correct on the source task but wrong on the target. The reading is that gain adaptation trained in a uniform-optimum regime should transfer poorly to a task whose closed-loop optimum is multimodal unless the rule is context-conditioned — a testable computational hypothesis that the present transfer slice illustrates within a single simulated task family.

Oracle-supervised gain as diagnostic upper bound. Appendix C reports an oracle-supervised correction-gain predictor that requires per-cell K-sweep labels. The closed-loop-oracle variant outputs $K \approx 0.02$ and matches K=0 on the focused grid within 1.5 cm at delay-18 cells, which is the strongest non-agent-available baseline for the correction-gain problem. The C2 single-cell SPSA result closes most of the way ($\sim$62% of the default-reliability vs K=0 gap, leaving $\sim$9 cm residual at n=200) toward this level using only signals the agent has access to — prediction errors during the reach and the reaching outcome afterward. We frame the oracle-supervised predictor not as a biological model but as the upper bound against which agent-available adaptation should be measured. The fact that a 10-parameter SPSA on per-episode outcome closes the gap on a single training cell suggests the upper bound is reachable by plausible biological signals; whether it is reachable at the population level (i.e. a single β over many cells) appears to depend on $K^\star_{\mathrm{cl}}$ structure as C3 shows.

Limits and open questions. (a) Two-layer rule only: β is the only learned object inside the reliability-adaptive observer; the forward model is fixed. A jointly-trained forward model + β pair is a natural extension. (b) Parameterisation ceiling: the feature-conditioned adapter of C4 conditions β on innovation statistics but retains the same logistic reliability-to-gain map $K_f = \sigma(\beta_{0,f} + \beta_{1,f} \log r_f)$, so the delay-0 ceiling at 0.71–0.73 m is unaffected. Alternative gain maps capable of representing $K_f \to 0$ (hardtanh, output clipping, or a direct K_f residual on top of K=0) are left to future work. (c) Two MyoSuite reach variants (ReachFixed and ReachRandom), 34-muscle arm only. Transfer to contact-rich tasks (grasp/place) and to other MyoSuite arms (Elbow/Finger reach) remains open. (d) Forward-model family: the residual MLP is a coarse instantiation of the cerebellar forward-model box; continuous-time recurrent alternatives such as LTC Hasani et al. (2021) and CfC Hasani et al. (2022) would provide a closer computational analogue when this observer is plugged into a full cortico-cerebellar loop. (e) The behaviour-cloned controller (Appendix C) washes out estimator differences at the demonstration scale we test; closed-loop state-estimation benchmarks should keep this in mind when choosing the downstream policy. (f) The oracle-supervised predictor of Appendix C is intentionally excluded from the main adaptive observer because it requires per-condition K-sweep labels the agent cannot generate online; it is reported only as a non-agent-available upper bound against which the within-trial + outcome adaptation of C2 can be measured.

6 Conclusion

We have presented a mechanistic decomposition of the closed-loop correction-gain geometry on a muscle-driven reaching task. The decomposition uses agent-available signals throughout — innovation history, per-episode reaching outcome, and z-scored log innovation statistics — and never accesses an oracle on the optimal gain.

Five claims combine. Default within-trial reliability is task- misaligned: innovation alone is not task-aware (C1). Single-cell SPSA on outcome closes most of that gap (C2). The closed-loop geometry is shaped by forward-model error and bias rather than by sensory reliability ($R^2 = 0.95$ vs. 0.17); a regression across four forward-model qualities makes the mechanism explicit (C3). Two distinct ceilings limit reliability-adaptive observers: a *context aggregation ceiling* (single global β cannot serve cells with different geometries) and a *parameterisation ceiling* (the reliability-to-gain logistic cannot represent the sharp $K{=}0$ regime that delay-0 cells require). A 30-parameter diagonal field-wise linear adapter on log innovation statistics breaks the context aggregation ceiling cleanly at $n{=}200$ episodes per cell — 6-cell-mean min-tip drops from 0.64 m to 0.56 m, with delay-18 cells reaching the $K{=}0$ baseline within 2.0 cm — while the parameterisation ceiling persists on delay-0 cells exactly as predicted (C4). Task transfer to the variable-target ReachRandom variant exhibits the geometry dependence directly: ReachFixed-trained β trails $K{=}0$ on ReachRandom delay-18 cells while default reliability is competitive there (C5).

We read this as a positive mechanistic result rather than a catalogue of limitations: *agent-available innovation statistics contain enough information in this focused-grid setting to overcome the context aggregation ceiling, and the residual delay-0 failure is a property of how reliability is mapped to a gain, not of the information available to the agent.* The parameterisation choice is therefore the substantive remaining design question for reliability-adaptive observers — a question this paper makes precise enough to answer in follow-up work.

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Statements and Declarations

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Competing interests. The author declares no competing interests.
Ethics approval. Not applicable. This study uses only the publicly available MyoSuite musculoskeletal simulation environment in MuJoCo; no human, animal, or clinical data are involved.
Consent to participate / consent for publication. Not applicable (no human subjects).
Data availability. All data used in this study are generated from the MyoSuite `myoArmReachFixed-v0` and `myoArmReachRandom-v0` environments Caggiano et al. (2022); generation scripts and configurations are included with the code release (Appendix G).
Code availability. The full reproduction pipeline (dataset generation, forward-model training, closed-loop evaluation, across-trial β adaptation, feature-conditioned β training, and figure generation) is released as open-source software at https://github.com/jkoba0512/myoarm-fse; an archived release DOI will be added upon acceptance. Frozen run identifiers required to regenerate every numerical claim and figure are listed in Appendix G.

Author contributions. Jun Kobayashi is the sole author and conducted all aspects of the work: conceptualisation, methodology, software, experiments, analysis, and manuscript preparation.

Appendix A Hyperparameters and Implementation Details

Table A1 lists every hyperparameter used in the paper. All training runs are reproducible from `configs/` in the source repository; estimator evaluation grids are listed verbatim in `configs/estimators/` and `configs/closed_loop/`.

Table A1 Hyperparameters used across the paper.

Component	Hyperparameter	Value
Forward model f_θ	state dim action dim hidden layers activation output parameters	83 34 2 × 256 ReLU, LayerNorm residual (Δx) 118,355
Forward training	optimiser learning rate weight decay batch size epochs val split	Adam Kingma and Ba 1×10^{-3} 0 256 200 (early stop patienc every 5th source episod
Multi-step loss	rollout horizon H	1, 4, 8
Stage A g_ϕ	input dim hidden layers output parameters	8 = 3 ctrl + 4 σ + 1 de 2 × 32 sigmoid scalar in $[0, 1]$ 1,409
Stage A training	optimiser learning rate weight decay epochs	Adam 1×10^{-3} 1×10^{-3} 500
Joint-PD controller	K_p, K_d action scale α muscle map	30, 3 5 $\mathrm{clip}(\alpha\,\mathrm{ReLU}(M\,u_{\mathrm{joint}})),$
IK solver	method max iter / tol damping λ^2	damped Jacobian pseuc 200 / 0.01 m 0.1
BC policy	state dim variants hidden layers output demos / training	83 (full) / 80 / 77 2 × 256 sigmoid 34-dim 30 ep. × 600 steps; 100
Eval grid (open-loop)	noise levels delays (steps) K values	none, low, medium, hig 0, 6, 18, 36 0.0, 0.25, 0.5, 0.75, 1.0
Eval grid (closed-loop)	noise levels (focused) delays (focused)	none, high, xhigh 0, 18

Appendix B Forward-Model Rollout Metrics

Table B2 reports horizon-wise prediction error for the three forward-model variants used in the paper. Multi-step rollout supervision substantially reduces medium- and long-horizon error while paying a small cost at h=1.

Table B2 Forward-model rollout MSE and tip prediction error on the held-out validation episodes. Lower is better. All three models share the same architecture and training set (51,534 transitions across 106 episodes); they differ only in the rollout horizon H.

	MSE			Tip err. (m)		
Model	h=1	h=10	h=50	h=1	h=10	h=50
H=1 (single-step)	0.0055	0.0124	0.0524	0.0076	0.0355	0.1381
H=4 multi-step	0.0053	0.0084	0.0164	0.0088	0.0247	0.0635
H=8 multi-step	0.0055	0.0082	0.0114	0.0098	0.0215	0.0483

Appendix C Oracle-Supervised Correction-Gain Predictor (Diagnostic Upper Bound)

The reliability-adaptive observer of Sec. 3.4 sets $K_f(t)$ from agent-available signals only. As a non-agent-available upper bound we also train an offline-oracle-supervised correction-gain predictor and evaluate it on the same grid; the single-cell SPSA result of C2 comes within ~ 9 cm of K=0 on its training cell, closing the bulk of the gap to this predictor.

Predictor architecture.

A small MLP $g_\phi : \mathbb{R}^8 \to [0, 1]$ predicts the scalar correction gain from a condition feature vector. The input

$$\xi = \left[\mathrm{onehot}(c)\,;\ \sigma\,;\ d/d_{\max}\right] \in \mathbb{R}^8, \qquad \text{(C1)}$$

concatenates a 3-way controller one-hot onehot(c), the 4-dim noise-sigma vector σ (`qpos`, `qvel`, `tip_pos`, `reach_err`), and a normalised delay $d/d_{\max}$ with $d_{\max}$=36 steps. The network is a $8 \to 32 \to 32 \to 1$ MLP with ReLU and a sigmoid output.

Oracles.

For each of the 72 conditions in the stress grid (3 controllers $\times$ 6 noise levels $\times$ 4 delays) we sweep $K \in K_{\text{grid}} = \{0, 0.25, 0.5, 0.75, 1.0\}$ and select the K minimising a chosen criterion. We define an *open-loop* oracle (estimation accuracy) and a *closed-loop* oracle (downstream task error):

$$K^\star_{\text{ol}}(\sigma, d, c) = \arg\min_{K \in K_{\text{grid}}} \mathbb{E}_t\big[\|\hat{x}_t(K) - x_t\|^2 \,\big|\, \sigma, d, c\big], \tag{C2}$$

$$K^\star_{\text{cl}}(\sigma, d, c) = \arg\min_{K \in K_{\text{grid}}} \mathbb{E}\big[\text{minTip}(K) \,\big|\, \sigma, d, c\big], \tag{C3}$$

where minTip is the closed-loop minimum tip-to-target distance over the episode. Two predictor variants g^{ol}_ϕ and g^{cl}_ϕ are trained against $K^\star_{\text{ol}}$ and $K^\star_{\text{cl}}$ respectively.

Closed-loop deployment.

Under the H=8 forward model on the focused 6-cell joint-PD grid, g^{ol}_ϕ outputs $K \approx 0.36$–0.99 and trails K=0 by 14–28 cm (the open-loop oracle never selects K=0 because the open-loop estimation error is minimised by trusting the observation when the buffer rollout is long). The closed-loop sweep of Appendix E fixes this: $K^\star_{\text{cl}} = 0$ at every cell, the retrained g^{cl}_ϕ outputs $K \approx 0.02$, and the predictor matches K=0 within 1.5 cm at the three delay-18 cells. A residual $\sim$ 17 cm gap persists at the three delay-0 cells because the closed-loop K-curve jumps sharply from K=0 to K=0.25 there (Appendix E) and the sigmoid asymptote 0.02 lies on the wrong side of the jump. Across the six cells, g^{cl}_ϕ closes 40–60% of the g^{ol}_ϕ-vs-K=0 gap with no architectural change to the predictor.

Transfer to the endpoint-error sensorimotor policy.

Repeating the closed-loop deployment with controller (iii) (endpoint-error sensorimotor policy, Sec. 3, item iii) shows the same qualitative pattern, in a weaker form: K=0 beats K=1 by -18.5 to -18.7 cm at the three d=0 cells and the two are within ± 3.2 cm at d=18 (Table C3). The endpoint-error policy's K=0 baseline is $\sim$ 11 cm worse in absolute terms than joint-PD's because the Jacobian-transpose projection is locally correct only; the qualitative pattern (closed-loop $K^\star = 0$ on this grid) is preserved.

Table C3 Oracle-supervised predictor deployed on the endpoint-error sensorimotor policy (iii) on the focused 6-cell grid under the H=8 forward model. Mean min-tip (m) over 10 episodes per cell. The "learned-cl" column is g^{cl}_ϕ trained on joint-PD $K^\star_{\text{cl}}$ labels.

noise	delay	K=0	K=1	learned-cl ($\bar{K} \approx 0.02$)
none	0	0.604	0.789	0.754
none	18	0.647	0.631	0.687
high	0	0.604	0.789	0.693
high	18	0.647	0.614	0.652
xhigh	0	0.604	0.791	0.656
xhigh	18	0.647	0.628	0.628

Behaviour-cloned policy washes out the signal.

The behaviour-cloned controller (iv) (a two-hidden-layer 256-unit sigmoid MLP trained on 30 scripted IK demonstrations) reaches better than the joint-PD baseline in absolute terms (success$_{<0.10\,\text{m}}$ from $\sim$ 12% to $\sim$ 30% in the easier cells) but the gap between any two estimators collapses to under 3 cm. We take this as evidence that the BC policy at this demonstration scale is effectively open-loop in the state channel and report it as a methodological caution for closed-loop estimator benchmarks: the downstream controller must be demonstrably state-coupled, or any observer differences will not surface. The behaviour-cloned controller is therefore retained only as a diagnostic and is not used in the main C1–C4 results.

Why we keep this as an upper-bound baseline only.

g^{cl}_ϕ requires a per-condition K-sweep that the biological agent does not have access to. We report it as the strongest baseline that uses non-agent-available information; the C2 single-cell SPSA result closes most of the gap to it on the training cell ($\sim$ 9 cm residual at n=200) while using only innovation history and per-episode outcome, both of which the agent can plausibly compute on-line.

Appendix D Full Open-Loop Oracle $K^\star$ Grids

Table D4 reports the mean oracle $K^\star$ per (delay, noise) cell under the H=8 multi-step-supervised forward model used throughout the main results, supervising the open-loop predictor variant g_ϕ^{ol} of Appendix C. Under single-step supervision (H=1) the $K^\star$<1 regime is confined to zero delay (18 of 72 cells); H=4 supervision widens the regime to 26 of 72 cells. The qualitative pattern (column-wise decrease at d=0, column-wise increase with delay) is the same across all three forward-model variants.

Table D4 Oracle correction gain $K^\star$ as a function of delay and noise under the H=8 multi-step-supervised forward model (mean over 3 controllers; 72 conditions total). 27 of 72 cells require $K^\star$<1.

	noise					
delay	none	low	medium	high	vhigh	xhigh
0	1.000	0.833	0.667	0.417	0.333	0.250
6	1.000	1.000	1.000	0.833	0.750	0.500
18	1.000	1.000	1.000	0.917	0.917	0.833
36	1.000	1.000	1.000	1.000	1.000	0.833

Appendix E Closed-Loop K-Sweep on the Focused Grid

Table E5 reports the closed-loop min-tip-to-target error per cell as a function of the (fixed) correction gain K, under the H=8 forward model and the joint-PD controller (10 episodes per cell). This K-sweep grounds two arguments in the main text: it defines the closed-loop oracle $K_{\mathrm{cl}}^\star$ used by the diagnostic predictor of Appendix C, and it is the baseline against which the reliability-adaptive observer is compared in Sec. 4.2–4.3. $K_{\mathrm{cl}}^\star$=0 in every cell of the focused grid.

Two regularities are worth noting. First, at zero delay the K-curve is sharp around K=0: moving from K=0 to K=0.25 jumps min-tip from 0.49 m to 0.77 m. The oracle-supervised predictor's sigmoid asymptote ($K\approx$0.025, Appendix C) sits on the wrong side of this jump in delay-0 cells, which is why the diagnostic gap remains ($\sim$ 17 cm). Second, at delay-18 the K-curve is smooth near K=0 (K=0 and K=0.25 within 0.01–0.05 m) so the predictor's small offset is harmless and the same predictor matches the K=0 baseline within 1.5 cm. These regularities also explain why the reliability-adaptive observer of C1 (which floats around $K_f \approx 0.6$–0.7) does badly at every delay-18 cell and at all delay-0 cells alike: any non-trivial K lands in the high-error plateau.

Table E5 Closed-loop min-tip error (m) per cell across the K-sweep (correction gain) under the H=8 forward model and the joint-PD controller. Each cell averages 10 episodes. Bold marks the per-cell minimum.

noise	delay	K=0	K=0.25	K=0.5	K=0.75	K=1
none	0	**0.493**	0.771	0.782	0.774	0.770
none	18	**0.497**	0.498	0.708	0.746	0.697
high	0	**0.493**	0.767	0.780	0.771	0.770
high	18	**0.497**	0.538	0.696	0.746	0.691
xhigh	0	**0.493**	0.772	0.769	0.766	0.762
xhigh	18	**0.497**	0.550	0.713	0.749	0.700

Appendix F Transfer Test on ReachRandom

Setup. We re-run a slim version of the pipeline on `myoArmReachRandom-v0` (same 34-muscle arm, same 600-step episodes, random per-episode target instead of a fixed one). Targets come from the same `runs/targets/train.npz` file (200 positions) so that targets are paired across estimators and forward models; initial joint pose is deterministic (q=0) for both variants. Transition data: 49,740 transitions across 106 episodes (random / lowamp / hold controllers), comparable to the 51,534- transition ReachFixed dataset. The marginal C2 cell-level effect ($\approx$ 8.6 cm at one cell with learned $K \approx 0.99$ and paired-t p=0.21) is task-dependent and *not* treated as a transfer target; the test asks whether the structural phenomena C1 / C3 / C4 persist under variable-target reaching. Pre-committed pass criteria, fixed before reading the numbers:

- *C1 replicate*: open-loop oracle $K_{\mathrm{ol}}^\star$ varies with noise / delay, with long-delay cells biasing toward K=1.
- *C3 replicate*: multi-step supervision reduces h=50 forward-model error and increases the

share of closed-loop cells where $K{=}0$ or prediction-heavy estimators beat $K{=}1$.

- *C4 replicate*: a closed-loop oracle $K^\star_{\text{cl}}$ trained predictor moves toward the K=0 / $K^\star_{\text{cl}}$ side of the K-curve compared with an open-loop oracle predictor.

Forward-model rollout error (Table F6) mirrors the ReachFixed pattern: $H{=}8$ multi-step rollout supervision reduces $h{=}50$ tip prediction error by 48% versus $H{=}1$ (0.110 → 0.057 m).

Table F6 ReachRandom forward-model rollout MSE and tip prediction error on held-out validation episodes. Same network and training recipe as the main text; lower is better.

	MSE			Tip err. (m)		
Model	$h{=}1$	$h{=}10$	$h{=}50$	$h{=}1$	$h{=}10$	$h{=}50$
$H{=}1$ (single-step)	0.0045	0.0103	0.0318	0.0066	0.0291	0.1098
$H{=}8$ multi-step	0.0046	0.0072	0.0096	0.0093	0.0229	0.0573

Closed-loop reaching, single-step model. Table F7 reports min-tip error at the 6-cell focused grid for $K{=}0$ and $K{=}1$ under the joint-PD controller. Two observations: (i) $K{=}0$ is constant across noise/delay because it ignores observations, and (ii) on ReachRandom $K{=}0$ already beats $K{=}1$ in *every* cell, by 7–27 cm. This contrasts with ReachFixed where the $H{=}1$ closed-loop optimum was nearer $K{=}1$ at delay-18 cells; the gap is therefore task-dependent at the single-step baseline.

Table F7 ReachRandom $H{=}1$ closed-loop min-tip error (m, mean over 10 episodes) under the joint-PD controller. $K{=}0$ beats $K{=}1$ in every cell by 7–27 cm.

noise	delay	$K{=}0$	$K{=}1$	$K{=}1-K{=}0$
none	0	0.499	0.770	+27.2 cm
none	18	0.498	0.578	+8.0 cm
high	0	0.499	0.770	+27.2 cm
high	18	0.498	0.567	+6.8 cm
xhigh	0	0.499	0.762	+26.3 cm
xhigh	18	0.498	0.593	+9.4 cm

Closed-loop K-curve, $H{=}8$ multi-step model. Table F8 reports the full $K \in \{0, 0.25, 0.5, 0.75, 1\}$ sweep at $H{=}8$. The closed-loop oracle $K^\star_{\text{cl}}$ is the per-cell minimiser. Unlike ReachFixed, where $K^\star_{\text{cl}}{=}0$ in every cell of the focused grid, on ReachRandom $K^\star_{\text{cl}}$ splits across three values: $K{=}0$ at all three $d{=}0$ cells (matching ReachFixed), $K{=}0.25$ at delay-18 in the low/medium-noise cells, and $K{=}1$ at the (xhigh, $d{=}18$) cell.

Table F8 ReachRandom $H{=}8$ closed-loop min-tip K-sweep (mean over 10 episodes). Bold marks the per-cell $K^\star_{\text{cl}}$.

noise	delay	$K{=}0$	$K{=}0.25$	$K{=}0.5$	$K{=}0.75$	$K{=}1$
none	0	**0.670**	0.762	0.785	0.780	0.770
none	18	0.642	**0.625**	0.706	0.644	0.647
high	0	**0.670**	0.764	0.785	0.775	0.770
high	18	0.642	**0.631**	0.688	0.651	0.639
xhigh	0	**0.670**	0.758	0.765	0.761	0.762
xhigh	18	0.642	0.662	0.691	0.650	**0.627**

Open-loop oracle structure on ReachRandom. We re-ran the full 3 controllers × 6 noise × 4 delay × 5 K stress eval (using 2 episodes per controller, matching the ReachFixed scale) for both forward models. $K^\star_{\text{ol}} < 1$ holds in 18 / 72 cells under $H{=}1$ (exactly matching ReachFixed at $H{=}1$) and 25 / 72 cells under $H{=}8$ (vs. 27 / 72 on ReachFixed); blending is again confined to delay-0 cells under $H{=}1$ and expands to delay ≥ 6 under $H{=}8$ (Table F9). C1 therefore replicates structurally on ReachRandom.

Table F9 ReachRandom $K^\star_{\text{ol}}$ (mean over 3 controllers) per (noise, delay) cell, under $H{=}1$ (top) and $H{=}8$ (bottom). Dashes mark $K^\star_{\text{ol}}{=}1.0$ (trust the observation). The non-trivial $K^\star_{\text{ol}} < 1$ regime is confined to delay-0 under $H{=}1$ and spreads to delay ≥ 6 under $H{=}8$, mirroring ReachFixed.

$H{=}1$ single-step forward model				
noise \ delay	0	6	18	36
none	–	–	–	–
low	0.83	–	–	–
medium	0.67	–	–	–
high	0.50	–	–	–
vhigh	0.33	0.92	–	–
xhigh	0.25	0.67	–	–
$H{=}8$ multi-step forward model				
noise \ delay	0	6	18	36
none	–	–	–	–
low	0.83	–	–	–
medium	0.67	–	–	–
high	0.42	0.92	–	–
vhigh	0.33	0.67	0.92	–
xhigh	0.25	0.50	0.75	0.75

Learned predictors trained on $K^\star_{\rm ol}$ vs $K^\star_{\rm cl}$. We train the same 8→32→32→1 Stage A predictor on (i) the $K^\star_{\rm ol}$ table from the H=8 ReachRandom stress eval, and (ii) the per-cell $K^\star_{\rm cl}$ labels from Table F8 (extended to the 72-cell schema by replicating the joint-PD label across the auxiliary controllers and the un-swept delays/noises). Table F10 reports the K each predictor outputs at the focused grid and the resulting closed-loop min-tip. The $K^\star_{\rm ol}$ predictor recovers values near K=1 at delay-18 cells (its training labels were $K^\star_{\rm ol}$=1 there), matching the K=1 baseline within 1 cm. The $K^\star_{\rm cl}$ predictor recovers small K (0.06–0.31) at delay-0 cells, beating the $K^\star_{\rm ol}$ predictor by 1.3–12.6 cm there. At delay-18 the $K^\star_{\rm cl}$ predictor's outputs drift toward intermediate values (K=0.24–0.95) and the resulting closed-loop reach is comparable to K=1 (within ±8 cm). The oracle-redefinition transfer therefore holds clearly at delay-0 but is muted at delay-18 because the closed-loop K-curve there is relatively flat (Table F8) and the sigmoid output cannot cleanly land on the mixed $K^\star_{\rm cl}$ (0/0.25/1) values across the three noise rows.

Table F10 ReachRandom H=8 closed-loop comparison of four estimators on the focused grid. Columns: K=0, K=1, *learned-ol* (Stage A trained on $K^\star_{\rm ol}$), *learned-cl* (Stage A trained on $K^\star_{\rm cl}$); for each learned predictor we report both its mean output $\bar{K}$ and the closed-loop min-tip (m, mean over 10 episodes).

noise	delay	K=0	K=1	learned-ol		learned-cl	
				$\bar{K}$	m	$\bar{K}$	m
none	0	0.670	0.770	0.80	0.781	0.06	0.655
none	18	0.642	0.647	0.98	0.638	0.24	0.673
high	0	0.670	0.770	0.71	0.777	0.09	0.690
high	18	0.642	0.639	0.96	0.644	0.47	0.721
xhigh	0	0.670	0.762	0.36	0.783	0.31	0.770
xhigh	18	0.642	0.627	0.88	0.654	0.95	0.642

Assessment against the pre-committed criteria.

- *C1 (full replication)*: open-loop oracle $K^\star_{\rm ol}$ varies with noise and delay in the same qualitative shape as ReachFixed, with the $K^\star_{\rm ol} < 1$ fraction growing from 18 / 72 (H=1) to 25 / 72 (H=8).
- *C3 (partial replication)*: H=8 rollout supervision reduces forward-model error by 48 % (replicates) and K=0 is optimal at all three delay-0 cells (replicates). The ReachFixed-style *regime shift* from K=1-winning at H=1 to K=0-winning at H=8 does *not* apply because K=0 was already winning at H=1; the variable-target variant is structurally more favourable to prediction-only across forward-model strengths.
- *C4 (partial replication, made richer)*: the closed-loop oracle $K^\star_{\rm cl}$ on ReachRandom is *not* uniformly K=0 (3 cells pick K=0, 2 cells pick K=0.25, 1 cell picks K=1), confirming $K^\star_{\rm cl}$ is a non-degenerate task-and-cell-dependent function. A Stage A predictor retrained on these labels matches the K=0 baseline within 1.3 cm at every delay-0 cell and beats the $K^\star_{\rm ol}$-trained predictor by 1–13 cm there. At delay-18, where $K^\star_{\rm cl}$ takes three different values across the three noise rows, the predictor's sigmoid output cannot cleanly land on each label, and the closed-loop benefit is muted.

Take-away. On ReachRandom the closed-loop oracle $K^\star_{\rm cl}$ is multimodal across cells ($\{0, 0.25, 1\}$), in contrast to the uniform $K^\star_{\rm cl} = 0$ on ReachFixed. The diagnostic oracle-supervised predictor $g^{\rm cl}_\phi$ of Appendix C transfers cleanly to delay-0 cells; at delay-18 the sigmoid-output parameterisation cannot fully exploit the cell-dependent $K^\star_{\rm cl}$. The same heterogeneity drives the C4 finding of Sec. 4.7: *default within-trial reliability* (no SPSA training of β) beats K=0 at delay-18 cells on ReachRandom because its $K_f \approx$ 0.6– 0.7 accidentally lands near the per-cell oracle values, while the ReachFixed-trained β pushes K_f globally lower and worsens performance on ReachRandom.

Appendix G Software, Hardware, and Reproducibility

Software stack.

Python 3.11, PyTorch (CPU), MuJoCo 3.x via Gymnasium 0.29, MyoSuite 2.12.2, NumPy / SciPy / pandas / matplotlib. Project-level dependencies are pinned in `pyproject.toml`.

Hardware.

All experiments run on a single CPU workstation (no GPU required). Forward-model training takes ∼30 min per model; closed-loop evaluation of 300 episodes (5 K-values × 6 cells × 10 episodes) takes ∼ 10 min. Per-cell SPSA training (100 iterations,

$S = 10$, 20 episodes per iteration) takes ~ 1 hour per cell when run alone; the 200-iteration feature-conditioned β training (Sec. 3.6) takes $\sim$3.5 hours.

Reproduction commands.

The following commands regenerate all numerical results and figures reported in the paper from the released repository. They assume `uv` is installed; substitute `python` otherwise.

Forward-model training.

```
# H=1, H=4, H=8 (ReachFixed expanded dataset)
uv run python scripts/train_forward_model.py \
  --config configs/models/mlp_expanded.yaml
uv run python scripts/train_forward_model.py \
  --config configs/models/mlp_expanded_multistep4.yaml
uv run python scripts/train_forward_model.py \
  --config configs/models/mlp_expanded_multistep8.yaml

# Undertrained H=8 (RQ2 degraded condition)
uv run python scripts/train_forward_model.py \
  --config configs/models/mlp_expanded_multistep8_undertrained.yaml
```

Closed-loop K-sweep across forward-model conditions (RQ2, Figs. 2, 6).

```
uv run python scripts/evaluate_closed_loop.py \
  --config configs/closed_loop/k_sweep_H1.yaml
uv run python scripts/evaluate_closed_loop.py \
  --config configs/closed_loop/k_sweep_H4.yaml
uv run python scripts/evaluate_closed_loop.py \
  --config configs/closed_loop/reliability_adaptive_v2_fullgrid.yaml
uv run python scripts/evaluate_closed_loop.py \
  --config configs/closed_loop/k_sweep_undertrained_H8.yaml
```

Forward-model diagnostics (Step 1 of RQ1).

```
uv run python scripts/compute_fm_diagnostics.py \
  --forward-models 2026-05-11T11-47-34Z \
                   2026-05-11T09-53-07Z \
                   2026-05-10T07-25-23Z \
                   2026-05-15T19-53-57Z \
  --n-episodes 3 \
  --output runs/diagnostics/geometry/fm_diag.csv
```

RQ1 regression (Sec. 4.5.1, Fig. 5).

```
uv run python scripts/rq1_geometry_regression.py
uv run python scripts/plot_rq1_geometry.py
```

RQ2 visualisation (Sec. 4.5.2, Fig. 6).

```
uv run python scripts/plot_rq2_fm_quality.py \
  --undertrained-h8-model 2026-05-15T19-53-57Z
```

Per-cell β diagnostic (Sec. 4.5.3, Table 2).

```
# 6 independent SPSA runs (~1 hour each at full CPU)
uv run python scripts/run_per_cell_spsa.py \
  --max-iter 100 --samples-per-side 10 \
  --a 2.0 --c 0.3 --A 5.0

# Fresh deployment eval (10 fresh episodes per cell)
uv run python scripts/eval_per_cell_beta.py
```

Feature-conditioned β adapter (Sec. 3.6, Sec. 4.6, Fig. 7).

```
# Pre-compute per-cell z-scored log innovation features
uv run python scripts/compute_per_cell_features.py \
  --n-warmup-episodes 5

# SPSA training (200 iter, S=12, ~3.5h walltime)
uv run python scripts/train_feature_conditioned_beta.py \
  --max-iter 200 --samples-per-cell 2 \
  --a 0.5 --c 0.3 --A 10 \
  --base global_spsa

# Fresh deployment eval (10 episodes per cell)
uv run python scripts/eval_feature_conditioned_beta.py

# F_adapt_compare figure (7-estimator comparison)
uv run python scripts/plot_rq3_adapt_compare.py
```

Other figures (existing reframe figures).

```
uv run python scripts/make_reframe_figures.py
# generates F_reliability_default, F_spsa_single, F_spsa_fullgrid
```

Repository layout.

```
src/myoarm_fse/
  envs/                MyoSuite wrappers, observation pipelines
  models/              ForwardMLP, training loops
  estimators/          FixedGainKalman, ReliabilityAdaptive, Learned
  controllers/         JointSpacePD, EndpointError, BC
  evaluation/          closed_loop.py (single episode rollout)
scripts/               all training / evaluation / plotting entries
configs/
  models/              forward-model training configs
  closed_loop/         closed-loop eval configs
  train/               SPSA / per-cell SPSA / feature-cond configs
runs/                  local outputs; large artifacts not versioned
figures/               committed PDF/PNG of paper figures
paper/                 LaTeX source + bibliography
```

Tests. Unit tests cover the per-step observer math, fixed-lag delay handling, and the deterministic build of every estimator. A MyoSuite-marked smoke suite verifies that one closed-loop episode runs end-to-end with each controller / estimator combination. Both suites pass on `main`.

```
uv run pytest              # default fast tests
uv run pytest -m myosuite # closed-loop integration smoke
```

Run artifacts.

The repository ships configs that, when run, write reproducible output directories under `runs/`. Each closed-loop or training run records its resolved config and a Git commit hash in `config.json`, so re-running with the same seed and commit exactly reproduces the headline numbers in this paper. Run identifiers used in the main results:

- Forward models: `2026-05-10T07-25-23Z` (H=1), `2026-05-11T09-53-07Z` (H=4), `2026-05-11T11-47-34Z` (H=8), `2026-05-15T19-53-57Z` (H=8 undertrained)
- Global SPSA β (Sec. 3.5): `2026-05-13T23-40-35Z`
- Per-cell β diagnostic (Sec. 4.5.3): `2026-05-15T12-59-01Z`
- Feature-conditioned β (Sec. 3.6): `2026-05-17T01-20-04Z`

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